Project: Worked examples
Title: Press and shrink fit
Status: DRAFT Author: TempestOS (2026-10-23) Checker: — (—) Issued: —

1. Scope and method

A shaft (solid or hollow) pressed or shrunk into a hub: the interface pressure for a given diametral interference, the hoop and radial stresses at the interface in hub and shaft, von Mises utilisation against each yield strength, the axial assembly force and the torque the joint can carry by friction, and the temperature rise to shrink-fit the hub.

Radial interference $\delta_r = \delta/2$. $p = \delta_r / [(R/E_h)\cdot((r_o^2 + R^2)/(r_o^2 - R^2) + \nu_h) + (R/E_s)\cdot((R^2 + r_i^2)/(R^2 - r_i^2) - \nu_s)]$. Hub bore: $\sigma_\theta = p\cdot(r_o^2 + R^2)/(r_o^2 - R^2)$, $\sigma_r = -p$. Shaft surface: $\sigma_\theta = -p\cdot(R^2 + r_i^2)/(R^2 - r_i^2)$, $\sigma_r = -p$. $F = \mu\cdot p\cdot 2\pi R\cdot L$; $T = \mu\cdot p\cdot 2\pi R^2\cdot L$; $\Delta T = (\delta + c_a)/(\alpha\cdot 2R)$.

Ref: Shigley's Mechanical Engineering Design §3-16, §3-17 — Press and shrink fits; stresses in pressurised cylinders
Ref: Roark's Formulas for Stress and Strain Ch. 13, Table 13.5 — Thick-walled cylinders under pressure

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
d	Interface (nominal) diameter	50	mm	drawing: DRG-HUB-04 rev C	1 … 1000 mm
delta	Diametral interference	40	µm	drawing: DRG-HUB-04 rev C	1 … 500 µm
d_o	Hub outer diameter	100	mm	drawing: DRG-HUB-04 rev C	1 … 2000 mm
d_i	Shaft bore diameter (0 for a solid shaft)	25	mm	drawing: DRG-SHAFT-13 rev A	0 … 1000 mm
L	Engagement length	60	mm	drawing: DRG-HUB-04 rev C	1 … 2000 mm
E_h	Hub Young's modulus	207	GPa	spec: steel	10 … 450 GPa
nu_h	Hub Poisson's ratio	0.292		spec: steel	0.1 … 0.45
E_s	Shaft Young's modulus	207	GPa	spec: steel	10 … 450 GPa
nu_s	Shaft Poisson's ratio	0.292		spec: steel	0.1 … 0.45
mu	Coefficient of friction at the interface	0.15		assumption: Dry steel on steel.	0.05 … 0.4
S_y_h	Hub yield strength	355	MPa	spec: EN 10025 S355	50 … 2000 MPa
S_y_s	Shaft yield strength	355	MPa	spec: EN 10025 S355	50 … 2000 MPa
alpha_h	Hub thermal expansion coefficient	0.000012	1/K	spec: steel	0.000005 … 0.00003 1/K
c_a	Assembly clearance wanted for shrink fitting	20	µm	assumption: Assembly clearance.	0 … 500 µm

3. Assumptions

A1. Both parts are long cylinders in plane stress; the interface pressure is uniform along the engagement. (Lamé thick-cylinder relations; end effects neglected.)
A2. The friction coefficient is as entered and the same for assembly and for torque transmission. (Enter it with its source; it dominates the force and torque results.)
A3. Yield is checked by the von Mises equivalent stress at the hub bore and at the shaft surface. (Axial stress taken as zero.)
A4. mu (Coefficient of friction at the interface) is taken as 0.15. (Dry steel on steel.)
A5. c_a (Assembly clearance wanted for shrink fitting) is taken as 20 µm. (Assembly clearance.)

4. Working

[15] R = d / 2
= 50 mm / 2
= 25 mm
Interface radius

[16] r_o = d_o / 2
= 100 mm / 2
= 50 mm
Hub outer radius

[17] r_i = d_i / 2
= 25 mm / 2
= 12.5 mm
Shaft bore radius

[18] delta_r = delta / 2
= 40 µm / 2
= 20 µm
Radial interference

[19] C_h = (R / E_h) * ((r_o^2 + R^2) / (r_o^2 - R^2) + nu_h) — Shigley §3-16
= 25 mm / 207 GPa × (((50 mm)^2 + (25 mm)^2) / ((50 mm)^2 - (25 mm)^2) + 0.292)
= 0.23655 mm/GPa
Hub compliance

[20] C_s = (R / E_s) * ((R^2 + r_i^2) / (R^2 - r_i^2) - nu_s) — Shigley §3-16
= 25 mm / 207 GPa × (((25 mm)^2 + (12.5 mm)^2) / ((25 mm)^2 - (12.5 mm)^2) - 0.292)
= 0.16602 mm/GPa
Shaft compliance

[21] p = delta_r / (C_h + C_s) — Shigley eq. 3-56
= 20 µm / (0.23655 mm/GPa + 0.16602 mm/GPa)
= 49.68 MPa
Interface pressure

[22] sigma_t_hub = p * (r_o^2 + R^2) / (r_o^2 - R^2) — Shigley eq. 3-57
= 49.68 MPa × ((50 mm)^2 + (25 mm)^2) / ((50 mm)^2 - (25 mm)^2)
= 82.8 MPa
Hub hoop stress at the bore

[23] sigma_r_hub = -p
= -49.68 MPa
= -49.68 MPa
Hub radial stress at the bore

[24] sigma_t_shaft = -p * (R^2 + r_i^2) / (R^2 - r_i^2) — Shigley eq. 3-58
= -49.68 MPa × ((25 mm)^2 + (12.5 mm)^2) / ((25 mm)^2 - (12.5 mm)^2)
= -82.8 MPa
Shaft hoop stress at the surface

[25] sigma_r_shaft = -p
= -49.68 MPa
= -49.68 MPa
Shaft radial stress at the surface

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[26] sigma_vm_hub = sqrt(sigma_t_hub^2 - sigma_t_hub * sigma_r_hub + sigma_r_hub^2)
= sqrt((82.8 MPa)^2 - 82.8 MPa × (-49.68 MPa) + (-49.68 MPa)^2)
= 115.92 MPa
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Hub von Mises stress at the bore
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[27] sigma_vm_shaft = sqrt(sigma_t_shaft^2 - sigma_t_shaft * sigma_r_shaft + sigma_r_shaft^2)
= sqrt((-82.8 MPa)^2 - (-82.8 MPa) × (-49.68 MPa) + (-49.68 MPa)^2)
= 72.183 MPa
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Shaft von Mises stress at the surface
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[28] F_a = mu * p * 2 * pi * R * L
= 0.15 × 49.68 MPa × 2 × π × 25 mm × 60 mm
= 70.233 kN
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Axial assembly (press) force
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[29] T = mu * p * 2 * pi * R^2 * L
= 0.15 × 49.68 MPa × 2 × π × (25 mm)^2 × 60 mm
= 1755.8 N·m
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Torque capacity by friction
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[30] dT = (delta + c_a) / (alpha_h * d)
= (40 µm + 20 µm) / (0.000012 1/K × 50 mm)
= 100 K
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Hub temperature rise for shrink fitting

5. Results

SYMBOL	VALUE	DESCRIPTION
p	49.68 MPa	Interface pressure
sigma_t_hub	82.8 MPa	Hub hoop stress at bore
sigma_t_shaft	-82.8 MPa	Shaft hoop stress at surface
sigma_vm_hub	115.92 MPa	Hub von Mises stress
sigma_vm_shaft	72.183 MPa	Shaft von Mises stress
F_a	70.233 kN	Assembly force
T	1755.8 N·m	Torque capacity
dT	100 K	Shrink-fit temperature rise

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
PASS	Hub yield	115.92 MPa	355 MPa	0.327	
PASS	Shaft yield	72.183 MPa	355 MPa	0.203	

7. Applicability

No warnings.

8. Verification

Template mech.press-fit@1.0.0 Content hash 45cae33eae60ff08 Engine calc-engine/1.0.0
Known-answer cases passed 2/2 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0